INTRODUCTION TO LATEX

Phan Thanh Hien et al.

University of Science and Technology of Hanoi

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Outline

1 Introduction

2 Introducing Overleaf.com

Did you know...

- Most of scientific research papers are prepared by LaTeX



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Studying Magnetic Fields and Dust in M17 Using Polarized Thermal Dust Emission Observed by SOFIA/HAWC+

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Abstract

We report on the highest spatial resolution measurement to date of magnetic fields (B-fields) in M17 using thermal dust polarization measurements taken by SOFIA/HAWC+ centered at a wavelength of 154 μm. Using the Davis–Chandrasekhar–Fermi method, in which the polarization angle dispersion calculated using the structure function technique is the quantity directly observed by SOFIA/HAWC+, we found the presence of strong B-fields of 980 ± 230 and 1665 ± 885 μG in the lower-density M17-N and higher-density M17-S regions, respectively. The B-field morphology in M17-N possibly mimics the fields in gravitationally collapsing molecular cores, while in M17-S the fields run perpendicular to the density structure. M17-S also displays a pillar feature and an asymmetric large-scale hourglass-shaped field. We use the mean B-field strengths to determine Alfvénic Mach numbers for both regions, finding that B-fields dominate over turbulence. We calculate the mass-to-flux ratio, λ, finding $\lambda = 0.07$ for M17-N and 0.28 for M17-S. These subcritical λ values are consistent with the lack of massive stars formed in M17. To study dust physics, we analyze the relationship between dust polarization fraction, p, emission intensity, I, gas column density, $N(H_2)$, polarization angle dispersion function, S, and dust temperature, T_d. p decreases with intensity as $I^{-\alpha}$ with $\alpha = 0.51$. p tends to first increase with T_d, but then decreases at higher T_d. The latter feature, seen in M17-N at high T_d when $N(H_2)$ and S decrease, is evidence of the radiative torque disruption effect.

Unified Astronomy Thesaurus concepts: Interstellar magnetic fields (845); Astrophysical magnetism (102); Magnetic fields (994); Star forming regions (1565); Stellar winds (1636); Star formation (1569); Interstellar dust (836); Interstellar dust processes (838)

1. Introduction

Star formation is a complex process that involves self-gravity, turbulence, magnetic fields, and stellar feedback. Understanding the exact role of magnetic fields in the evolution of molecular clouds (MCs) and in the star formation process is a challenge of modern astrophysics. In the past decades, evidence has emerged to suggest the importance of magnetic fields (B-fields) in the evolution of MCs and star formation (McKee & Ostriker 2007; Ward-Thompson et al. 2020). There are two types of B-field models, in which B-fields play contrasting roles. First, strong B-field models support a paradigm in which magnetic pressure acts against the gravitational collapse of MCs. This magnetic support is lost when the ratio of the core mass to the magnetic flux exceeds a critical value, beyond which the cloud enters into a state of gravitational collapse to form a new star (Nakano & Nakamura 1978). Second, in the weak-field models, B-fields are sufficiently weak that the dynamics of MCs are dominated by turbulence. Star formation takes place in filaments that are likely produced at the intersection of supersonic turbulent flows (e.g., Padoan & Nordlund 1999; Elmegreen 2000; Mac Low & Klessen 2004; Crutcher 2012). It is thus necessary to observe B-fields in specific MCs (e.g., M17 in this work) to investigate their effects on star formation and on the subsequent evolution of the entire cloud, in order to test these theoretical predictions (Seifried & Walch 2015; Federrath 2016; Li & Klein 2019; Ward-Thompson et al. 2020).

Dust polarization arising from aligned dust grains is widely used to map B-fields (see, e.g., Crutcher 2012). This relies on thermal dust emission being preferentially polarized perpendicular to the local B-field direction. This assumption is valid in most astrophysical environments (e.g., the diffuse interstellar medium (ISM) and MCs) where grain alignment occurs with the shortest axis parallel to the magnetic field (see

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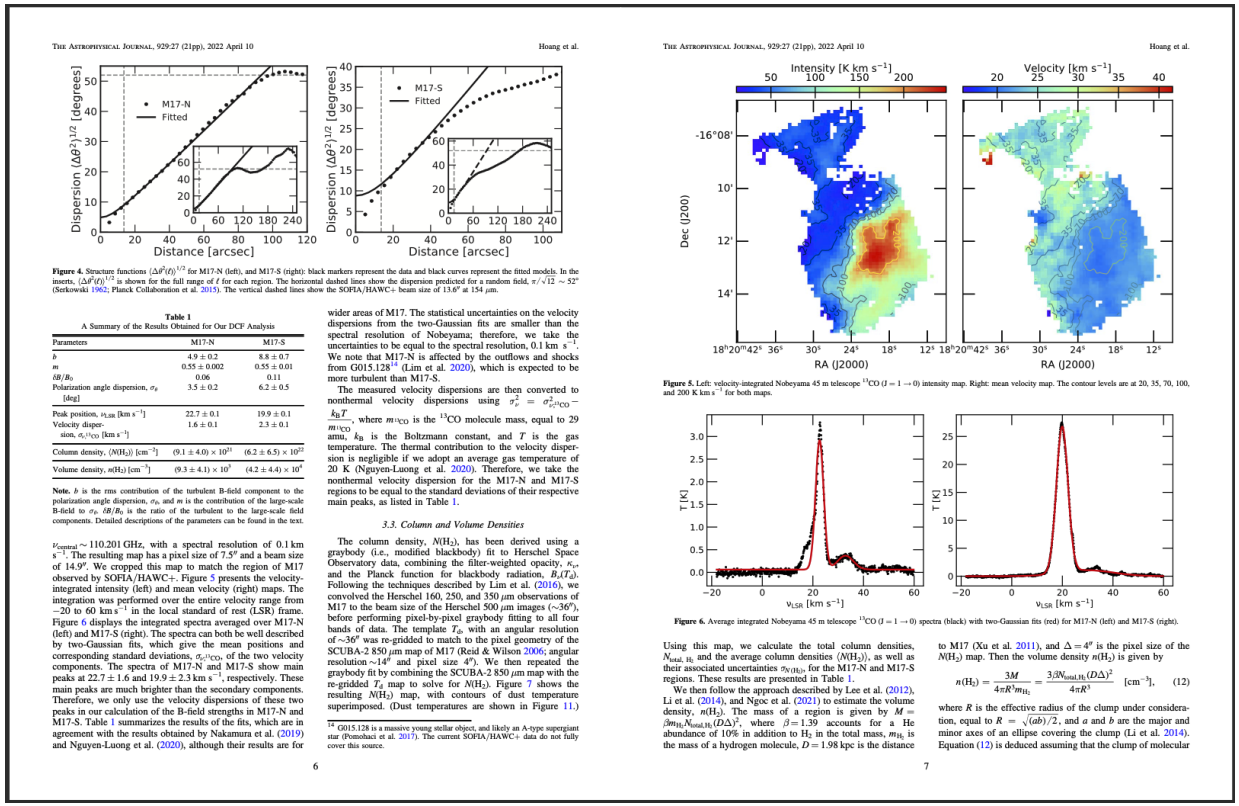


Figure 4. Structure functions $\langle \Delta\theta^2(\ell) \rangle^{1/2}$ for M17-N (left), and M17-S (right): black markers represent the data and black curves represent the fitted models. In the inserts, $\langle \Delta\theta^2(\ell) \rangle^{1/2}$ is shown for the full range of ℓ for each region. The horizontal dashed lines show the dispersion predicted for a random field, $\pi/\sqrt{12} \sim 52°$ (Serkowski 1962; Planck Collaboration et al. 2015). The vertical dashed lines show the SOFIA/HAWC+ beam size of 13.6″ at 154 μm.

Table 1
A Summary of the Results Obtained for Our DCF Analysis

Parameters	M17-N	M17-S
b	4.9 ± 0.2	8.8 ± 0.7
m	0.55 ± 0.002	0.55 ± 0.01
$\delta B/B_0$	0.06	0.11
Polarization angle dispersion, σ_θ [deg]	3.5 ± 0.2	6.2 ± 0.5
Peak position, $v_{\rm LSR}$ [km s^{-1}]	22.7 ± 0.1	19.9 ± 0.1
Velocity dispersion, σ_v^{13}CO [km s^{-1}]	1.6 ± 0.1	2.3 ± 0.1
Column density, $\langle N(H_2) \rangle$ [cm^{-2}]	$(9.1 \pm 4.0) \times 10^{21}$	$(6.2 \pm 6.5) \times 10^{22}$
Volume density, n(H$_2$) [cm^{-3}]	$(9.3 \pm 4.1) \times 10^3$	$(4.2 \pm 4.4) \times 10^4$

Note. b is the rms contribution of the turbulent B-field component to the polarization angle dispersion, σ_θ, and m is the contribution of the large-scale B-field to σ_θ. $\delta B/B_0$ is the ratio of the turbulent to the large-scale field components. Detailed descriptions of the parameters can be found in the text.

$\nu_{\rm central} \sim 110.201$ GHz, with a spectral resolution of 0.1 km s^{-1}. The resulting map has a pixel size of 7.5″ and a beam size of 14.9″. We cropped this map to match the region of M17 observed by SOFIA/HAWC+. Figure 5 presents the velocity-integrated intensity (left) and mean velocity (right) maps. The integration was performed over the entire velocity range from −20 to 60 km s^{-1} in the local standard of rest (LSR) frame. Figure 6 displays the integrated spectra averaged over M17-N (left) and M17-S (right). The spectra can both be well described by two-Gaussian fits, which give the mean positions and corresponding standard deviations, σ_v^{13}CO, of the two velocity components. The spectra of M17-N and M17-S show main peaks at 22.7 ± 1.6 and 19.9 ± 2.3 km s^{-1}, respectively. These main peaks are much brighter than the secondary components. Therefore, we only use the velocity dispersions of these two peaks in our calculation of the B-field strengths in M17-N and M17-S. Table 1 summarizes the results of the fits, which are in agreement with the results obtained by Nakamura et al. (2019) and Nguyen-Luong et al. (2020), although their results are for wider areas of M17. The statistical uncertainties on the velocity dispersions from the two-Gaussian fits are smaller than the spectral resolution of Nobeyama; therefore, we take the uncertainties to be equal to the spectral resolution, 0.1 km s^{-1}. We note that M17-N is affected by the outflows and shocks from G015.128[14] (Lim et al. 2020), which is expected to be more turbulent than M17-S.

The measured velocity dispersions are then converted to nonthermal velocity dispersions using $\sigma_{\rm nt}^2 = \sigma_{v,^{13}\rm CO}^2 - \frac{k_B T}{m_{^{13}\rm CO}}$, where $m_{^{13}\rm CO}$ is the ^{13}CO molecule mass, equal to 29 amu, k_B is the Boltzmann constant, and T is the gas temperature. The thermal contribution to the velocity dispersion is negligible if we adopt an average gas temperature of 20 K (Nguyen-Luong et al. 2020). Therefore, we take the nonthermal velocity dispersion for the M17-N and M17-S regions to be equal to the standard deviations of their respective main peaks, as listed in Table 1.

3.3. Column and Volume Densities

The column density, $N(H_2)$, has been derived using a graybody (i.e., modified blackbody) fit to Herschel Space Observatory data, combining the filter-weighted opacity, κ_{ν_i}, and the Planck function for blackbody radiation, $B_\nu(T_d)$. Following the techniques described by Lim et al. (2016), we convolved the Herschel 160, 250, and 350 μm observations of M17 to the beam size of the Herschel 500 μm images (∼36″), before performing pixel-by-pixel graybody fitting to all four bands of data. The template T_d, with an angular resolution of ∼36″ was re-gridded to match to the pixel geometry of the SCUBA-2 850 μm map of M17 (Reid & Wilson 2006; angular resolution ∼14″ and pixel size 4″). We then repeated the graybody fit by combining the SCUBA-2 850 μm map with the re-gridded T_d map to solve for $N(H_2)$. Figure 7 shows the resulting $N(H_2)$ map, with contours of dust temperature superimposed. (Dust temperatures are shown in Figure 11.)

[14] G015.128 is a massive young stellar object, and likely an A-type supergiant star (Pomohaci et al. 2017). The current SOFIA/HAWC+ data do not fully cover this source.

Figure 5. Left: velocity-integrated Nobeyama 45 m telescope ^{13}CO ($J = 1 \rightarrow 0$) intensity map. Right: mean velocity map. The contour levels are at 20, 35, 70, 100, and 200 K km s^{-1} for both maps.

Figure 6. Average integrated Nobeyama 45 m telescope ^{13}CO ($J = 1 \rightarrow 0$) spectra (black) with two-Gaussian fits (red) for M17-N (left) and M17-S (right).

Using this map, we calculate the total column densities, $N_{\rm total,\,H_2}$ and the average column densities $\langle N(H_2) \rangle$, as well as their associated uncertainties $\sigma_{\langle N(H_2) \rangle}$, for the M17-N and M17-S regions. These results are presented in Table 1.

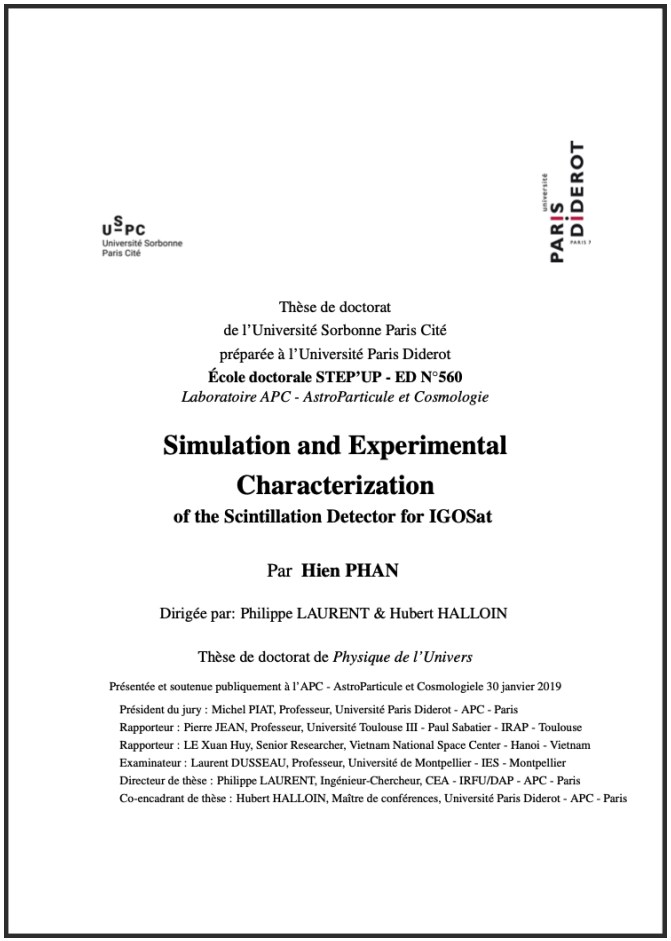
We then follow the approach described by Lee et al. (2012), Li et al. (2014), and Ngoc et al. (2021) to estimate the volume density, $n(H_2)$. The mass of a region is given by $M = \beta m_{\rm H} N_{\rm total,\,H_2} (D\Delta)^2$, where $\beta = 1.39$ accounts for a He abundance of 10% in addition to H_2 in the total mass, $m_{\rm H}$ is the mass of a hydrogen molecule, $D = 1.98$ kpc is the distance to M17 (Xu et al. 2011), and $\Delta = 4''$ is the pixel size of the $N(H_2)$ map. Then the volume density $n(H_2)$ is given by

$$n(\mathrm{H_2}) = \frac{3M}{4\pi R^3 m_{\rm H_2}} = \frac{3\beta N_{\rm total,\,H_2} (D\Delta)^2}{4\pi R^3} \quad [\mathrm{cm^{-3}}], \qquad (12)$$

where R is the effective radius of the clump under consideration, equal to $R = \sqrt{(ab)/2}$, and a and b are the major and minor axes of an ellipse covering the clump (Li et al. 2014). Equation (12) is deduced assuming that the clump of molecular

Did you know...

- LaTeX is also using for scientific reports, books, etc.



Thèse de doctorat
de l'Université Sorbonne Paris Cité
préparée à l'Université Paris Diderot
École doctorale STEP'UP - ED N°560
Laboratoire APC - AstroParticule et Cosmologie

Simulation and Experimental Characterization
of the Scintillation Detector for IGOSat

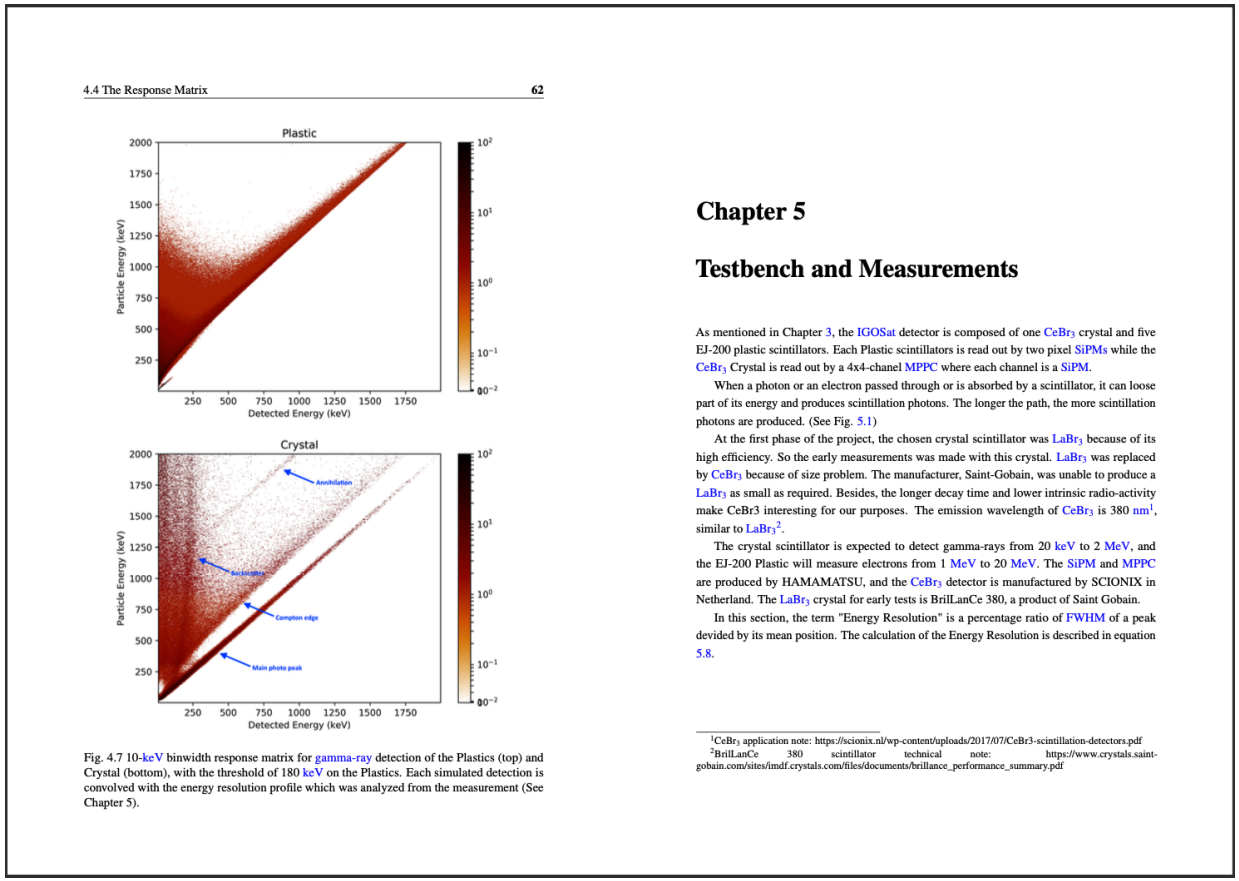
Par **Hien PHAN**

Dirigée par: Philippe LAURENT & Hubert HALLOIN

Thèse de doctorat de *Physique de l'Univers*

Présentée et soutenue publiquement à l'APC - AstroParticule et Cosmologiele 30 janvier 2019

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Co-encadrant de thèse : Hubert HALLOIN, Maître de conférences, Université Paris Diderot - APC - Paris



4.4 The Response Matrix 62

Fig. 4.7 10-keV binwidth response matrix for gamma-ray detection of the Plastics (top) and Crystal (bottom), with the threshold of 180 keV on the Plastics. Each simulated detection is convolved with the energy resolution profile which was analyzed from the measurement (See Chapter 5).

Chapter 5

Testbench and Measurements

As mentioned in Chapter 3, the IGOSat detector is composed of one $CeBr_3$ crystal and five EJ-200 plastic scintillators. Each Plastic scintillators is read out by two pixel SiPMs while the $CeBr_3$ Crystal is read out by a 4x4-chanel MPPC where each channel is a SiPM.

When a photon or an electron passed through or is absorbed by a scintillator, it can loose part of its energy and produces scintillation photons. The longer the path, the more scintillation photons are produced. (See Fig. 5.1)

At the first phase of the project, the chosen crystal scintillator was $LaBr_3$ because of its high efficiency. So the early measurements was made with this crystal. $LaBr_3$ was replaced by $CeBr_3$ because of size problem. The manufacturer, Saint-Gobain, was unable to produce a $LaBr_3$ as small as required. Besides, the longer decay time and lower intrinsic radio-activity make CeBr3 interesting for our purposes. The emission wavelength of $CeBr_3$ is 380 nm[1], similar to $LaBr_3$[2].

The crystal scintillator is expected to detect gamma-rays from 20 keV to 2 MeV, and the EJ-200 Plastic will measure electrons from 1 MeV to 20 MeV. The SiPM and MPPC are produced by HAMAMATSU, and the $CeBr_3$ detector is manufactured by SCIONIX in Netherland. The $LaBr_3$ crystal for early tests is BrilLanCe 380, a product of Saint Gobain.

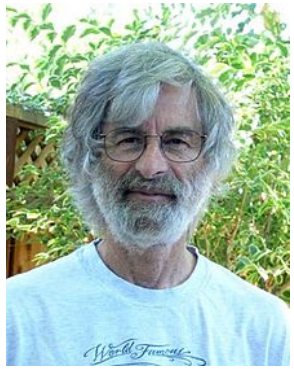
In this section, the term "Energy Resolution" is a percentage ratio of FWHM of a peak devided by its mean position. The calculation of the Energy Resolution is described in equation 5.8.

[1] $CeBr_3$ application note: https://scionix.nl/wp-content/uploads/2017/07/CeBr3-scintillation-detectors.pdf
[2] BrilLanCe 380 scintillator technical note: https://www.crystals.saint-gobain.com/sites/imdf.crystals.com/files/documents/brillance_performance_summary.pdf

A bit of history

- Late 1970s - 1989: TEX by Don Knuth
  - Decent typesetting by the authors!
  - Same output now as in the future
  - Low level, but powerful
  - Frozen development
- Starting early 1980s: LATEX by Leslie Lamport
  - High level language (macros, really) for TeX
  - Easy to use!
  - Current version is Latex2e





What is LaTeX?

- LaTeX is pronounced "lay-tech" or "lah-tech" not "la-teks"
- LaTeX is a document preparation system used for high-quality typesetting.
- It is most often used for medium-to-large technical or scientific documents but it can be used for almost any form of publishing.
- LaTeX markup describes the content and layout of the document, as opposed to the formatted text found in **WYSIWYG** word processors like Google Docs, LibreOffice Writer and Microsoft Word.



Figure: Logo of the LaTeX project

Why LaTeX?

- It makes beautiful documents
  - Especially mathematics
- It was created by scientists, for scientists
  - A large and active community
- It is powerful — you can extend it
  - Packages for papers, presentations, spreadsheets, . . .

How does it work?

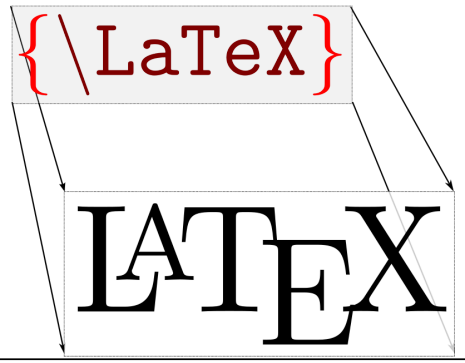
- You write your document in plain text with commands that describe its structure and meaning.
- The latex program processes your text and commands to produce a beautifully formatted document.

```
The rain in Spain falls \emph{mainly} on the plain.
```



The rain in Spain falls *mainly* on the plain.

Examples

\begin{itemize} \item Tea \item Coffee \item Milk \end{itemize}	■ Tea ■ Coffee ■ Milk
\begin{figure} \includegraphics{LaTeX_cover.png} \end{figure}	
\begin{equation} \alpha + \beta + 1 \end{equation}	$\alpha + \beta + 1 \quad (1)$

LaTeX has...

LaTeX is a sophisticated document preparation system. It has:

- Stylistic uniformity
- Bibliography support
- Sophisticated structuring abilities
- Reference tracking
- Highly extendable capabilities

LaTeX does not...

LaTeX is a not a word processor. So it does not:

- Spell-check your documents[1]
- Give you complete control over formatting
- Provide a graphical interface for editing

[1] You can use the third-party tools; Google Chrome can also check it for you.

LaTeX is not a word processor!

- Instead, LaTeX encourages authors not to worry too much about the appearance of their documents but to concentrate on getting the right content.
- To produce the document in the following box in most typesetting or word-processing systems, the author would have to decide, for example:
  - what layout to use
  - 18pt Times Roman for the title
  - 12pt Times Italic for the name...
- These tasks would lead to two results: authors wasting their time with designs; and a lot of badly designed documents!

```
Introduction to Informatics
Dr.  Phan Thanh Hien
October 2024
Hello world!
```

Introduction to Informatics

Dr. Phan Thanh Hien
October 2024

Hello world!

LaTeX uses commands

- LaTeX is based on the idea that *it is better to leave document design to document designers*, and to let authors get on with writing documents.

In LaTeX you would input this document as:

```
\documentclass{article}
\title{Introduction to
Informatics}
\author{Dr. Phan Thanh Hien}
\date{October 2024}

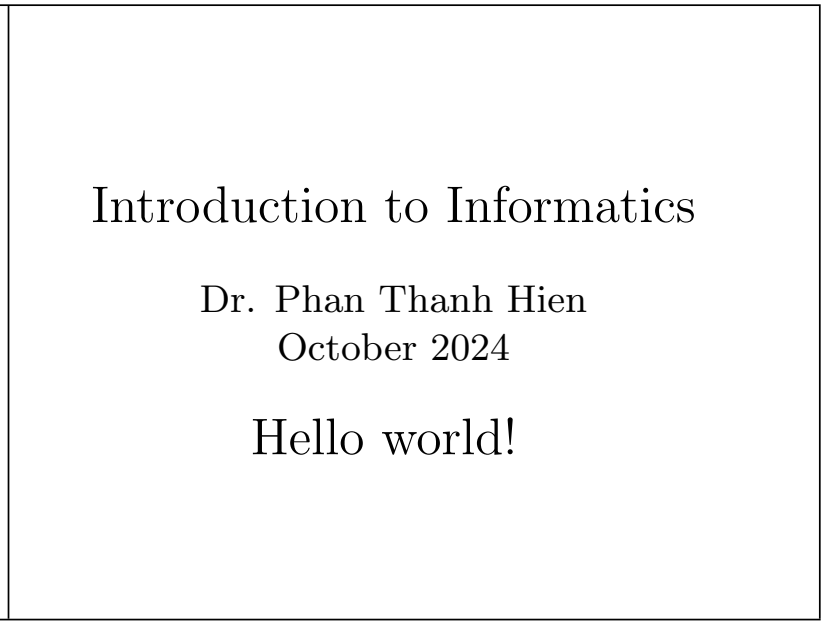
\begin{document}
\maketitle
Hello world! %your content goes
here...
\end{document}
```

- This document is an *article*.
- Its title is *Introduction to Informatics*.
- Its author is *Dr. Phan Thanh Hien*.
- It was written in *October 2024*.

- **The document starts here**
- Title and information will be inserted here.
- It is followed by the text *Hello world!*
- **The document ends here**

LaTeX concentrates on getting the right content

```
\documentclass{article}
\title{Introduction to
Informatics}
\author{Dr. Phan Thanh Hien}
\date{October 2024}
\begin{document}
\maketitle
Hello world! %your content goes
here...
\end{document}
```



- Commands start with a *backslash* \.
- Every document starts with a `\documentclass` command.
- The *argument* in curly braces { } tells LaTeX what kind of document we are creating: *an article*.
- A percent sign % starts a *comment* — LaTeX will ignore the rest of the line.

Example

```
\documentclass{article}
\usepackage{amsmath}
\title{\LaTeX}

\begin{document}

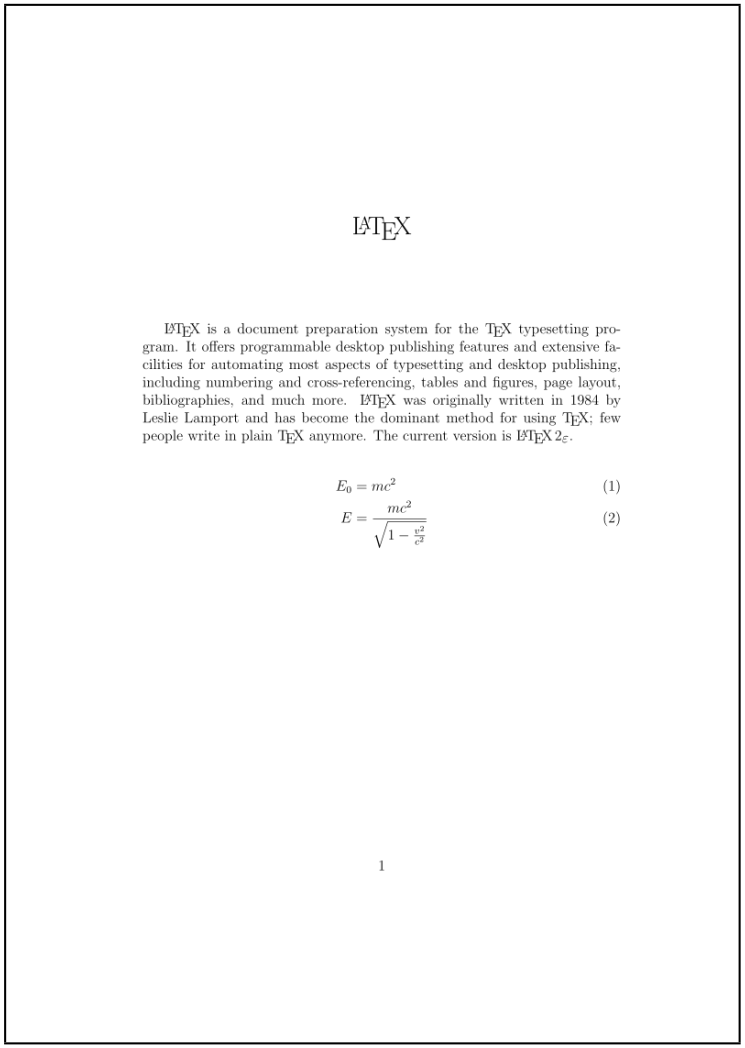
\maketitle

\LaTeX{} is a document preparation system for the \TeX{}
typesetting program. It offers programmable desktop publishing
features and extensive facilities for automating most aspects of
typesetting and desktop publishing, including numbering and
cross-referencing, tables and figures, page layout, bibliographies, and
much more. \LaTeX{} was originally written in 1984 by Leslie
Lamport and has become the dominant method for using \TeX; few
people write in plain \TeX{} anymore. The current version is
\LaTeXe{}.

% This is a comment, not shown in final output.
% The following shows typesetting power of LaTeX:

\begin{align}
   E_0 &= mc ^2
   E &= \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}
\end{align}

\end{document}
```



LaTeX

LaTeX is a document preparation system for the TeX typesetting program. It offers programmable desktop publishing features and extensive facilities for automating most aspects of typesetting and desktop publishing, including numbering and cross-referencing, tables and figures, page layout, bibliographies, and much more. LaTeX was originally written in 1984 by Leslie Lamport and has become the dominant method for using TeX; few people write in plain TeX anymore. The current version is LaTeX 2_ε.

$$E_0 = mc^2 \quad (1)$$

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (2)$$

1

Attitude adjustment

- Use commands to describe **'what it is'**, not *'how it looks'*.
- Focus on your content.
- Let LaTeX do its job.

LaTeX vs. Word processor

- With a word processor, you spend valuable time agonizing over what font size to make the section headings.
  With LaTeX, you just tell it to start a new section.
- With a word processor, changing the formatting means you have to change each instance individually.
  With LaTeX, you just redefine the relevant commands.
- With a word processor, you have to carefully match any provided templates.
  With LATEX, you can be sure you've fit the template, and switch templates easily.

LaTeX Features

- Typesetting journal articles, technical reports, books, and slide presentations.
- Control over large documents containing sectioning, cross-references, tables and figures.
- Typesetting of complex mathematical formulas.
- Advanced typesetting of mathematics with AMS-LaTeX.
- Automatic generation of bibliographies and indexes.
- Multi-lingual typesetting.
- Inclusion of artwork, and process or spot colour.
- Using PostScript or Metafont fonts.

Table of Contents

1 Introduction

2 Introducing Overleaf.com

What is Overleaf?



Write like a rocket scientist with Overleaf
—the collaborative, online LaTeX editor that anyone can use.

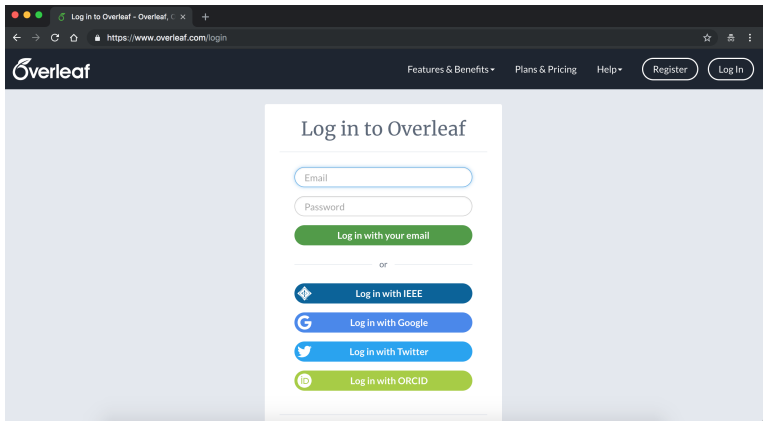
Website: Overleaf.com

- Overleaf is a startup and social enterprise that builds modern collaborative authoring tools to help make science and research faster, more open and more transparent.
- Overleaf's market-leading collaboration technology is now in use by over 18 million researchers, students, and teachers in institutions, labs, and industry worldwide.

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- To start using Overleaf go to www.overleaf.com.
- If you don't have an account enter your e-mail address and set a password in the corresponding boxes.
- If you already have an account, click Login in the upper right corner, then type in your email and password and click the Login button.



Once you've logged in already, you can start a new project and give a name to it: "Hello LaTeX"

The Editor interface

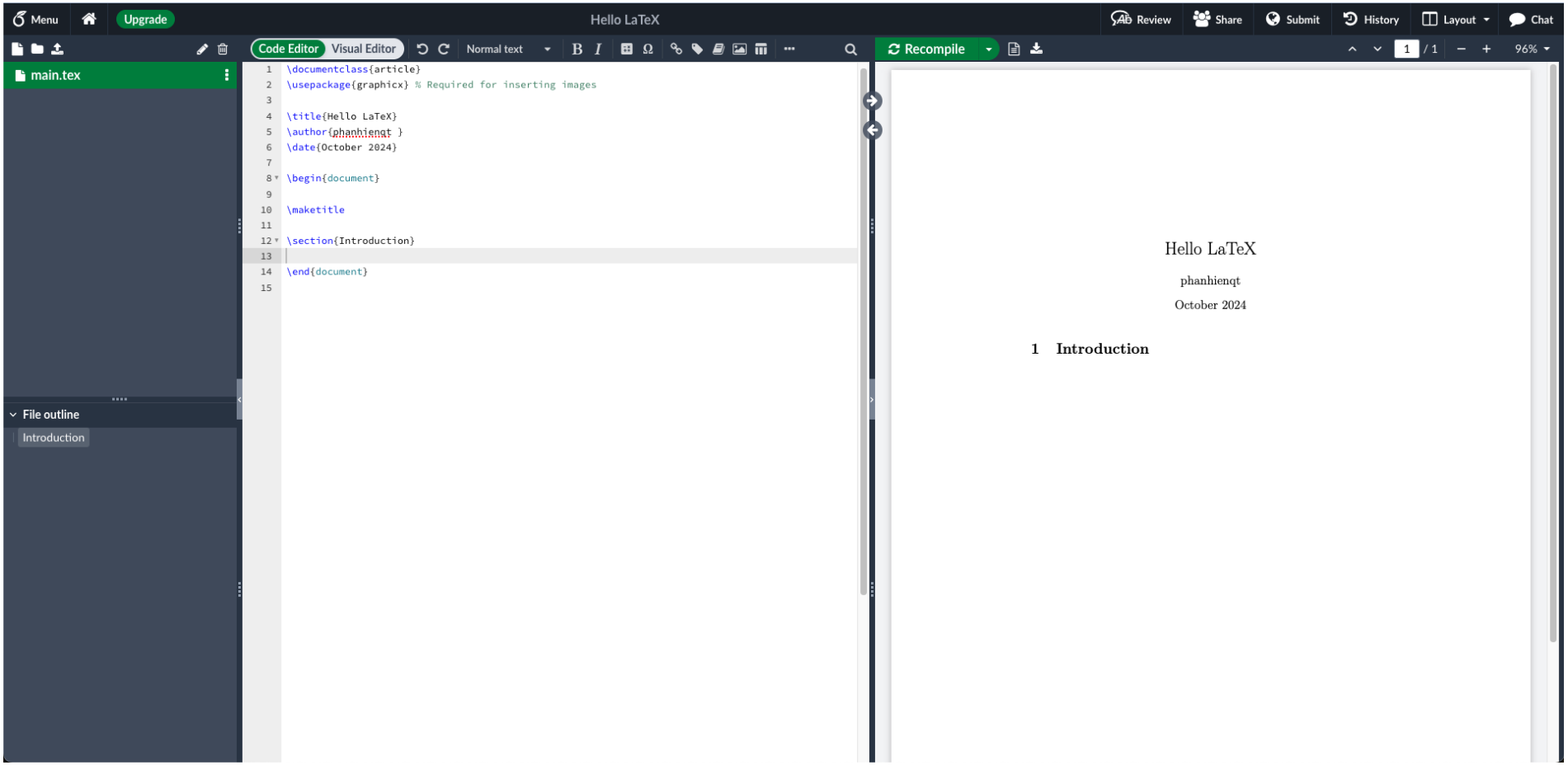
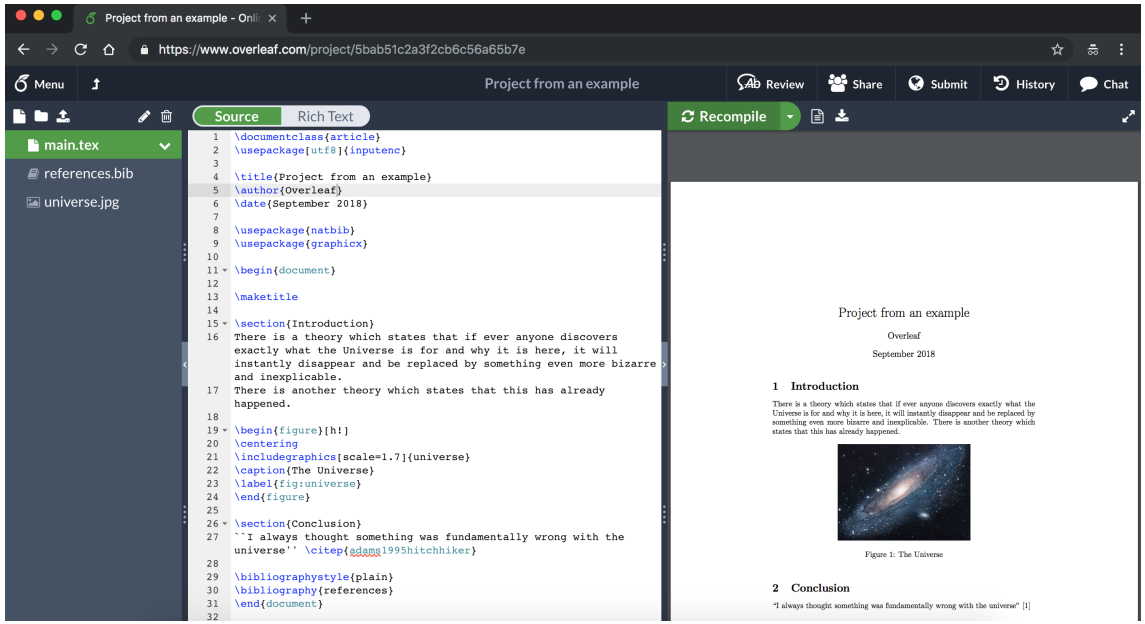


Figure: The Editor interface of the Overleaf with some basic information already filled in. You can start editing your .tex file now, to view the changes click Recompile.

Starting a new project

- In the Project Management page click **New project / Blank Project**, enter a name for your project then click Create.
- In the Project Management page click **New project / Example Project**, enter a name for your project then click Create.



Thank you!

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